

Dentfluence

Application Testing & Hardening — Final Report

Prepared for	Sumit (solo builder & dentist)
Date	23 June 2026
Scope	Whole-app, module-by-module test sweep + bug fixes
Stack	Laravel, MySQL (Laragon), Blade, Alpine.js

Executive summary

Every module of the Dentfluence app has now been put through an automated test sweep and a code audit. The goal was simple: make sure buttons actually work, screens don't silently break, and the hidden automations (reminders, stock updates, accounting entries) fire correctly — without you having to click through the whole app by hand after every change.

The result is a three-layer safety net plus a batch of real bugs fixed along the way — including two that were quietly broken in your live app (appointment **Revert** and **cancel reason**). From here on, if a future change breaks something, a test will catch it instead of a patient-facing surprise.

11 modules swept	~32 automated tests	8 real bugs fixed	1 nightly self-test
----------------------------	-------------------------------	-----------------------------	-------------------------------

The three-layer safety net

Each layer catches a different kind of problem, so together they cover the app from the page level all the way down to the invisible automations.

Layer	What it does	Catches
1. Nightly crawler	Logs in and visits every page at 2am, then writes a colour-coded report.	Crashes, blank / truncated pages, broken reports
2. Browser tests (~20, Dusk)	Drive the real screens like a person — click buttons, fill forms, save.	Forms that don't save, broken daily-use flows
3. Feature tests (12)	Check the invisible chains — one action quietly triggering another.	Reminders, stock math, accounting, status logic

Modules covered

All 11 module areas were audited for code “drift” (stale column / status names) and given tests.

Module	What was tested
Finance	Drift audit + Dusk forms + automation tests
Lab	Drift audit (5 stale-status spots fixed) + Dusk + auto-close

Communication / PRM	Drift audit + Dusk + Feature tests
HR	Drift audit + Dusk + auto-absent test
Prescriptions	Drift audit + form tests
Inventory	Stock math + full PO→GRN→stock→payable chain
Treatments	Create + auto-draft-SOP automation
Clinical Library	Drift audit (file upload already covered)
Marketing	Idea → campaign conversion chain
Appointments	Status workflow + revert (2 live bugs fixed)
Settings	Masters save path (feeds every dropdown)

Bugs found & fixed

These were caught by the sweep. The ones marked **live** were silently affecting the running app.

Issue	What was wrong	Status
Appointments: Revert & cancel-reason	Two columns weren't "fillable", so Revert never worked and cancel reasons were lost.	Live · Fixed
Recall engine not queuing	Reminder cooldown fields were silently dropped, so recall reminders never queued.	Live · Fixed
Lab status drift (5 places)	Old status names lingered after the Lab v2 rebuild, breaking auto-close, recall & KPIs.	Live · Fixed
Finance KPI / analytics columns	"Closed cases" and analytics used wrong column names → wrong numbers.	Live · Fixed
21 broken pages	First crawler run flagged 21 pages erroring; traced to 5 root causes.	Fixed
Fresh-install migration blocker	A migration ordering issue would break a clean install.	Fixed
Duplicate scheduler entries	Two reminder tasks were registered twice → firing (and notifying) twice.	Fixed
Treatment-plan dropdown refresh	New/accepted plan only showed in the visit screen after a page reload.	Fixed

Test suite inventory

Feature tests (the invisible automations)

- Lab case auto-close
- Recall engine (lab received)
- Finance payment automation
- Task → notification

- HR auto-absent
- Inventory stock movement
- Inventory PO receive → GRN + stock + payable
- Inventory item create
- Treatment create → auto draft SOP
- Marketing idea → campaign
- Appointment status flow + revert
- Settings masters save

Browser tests (the daily-use screens)

Login, add patient, treatment plan, treatment visit, invoice, membership, wallet, documents, communication, notes, lab case, prescription, plus finance (expense / vendor / payroll), lab vendor, communication opportunity, HR training, and prescription drug forms — around 20 in total.

How to run & maintain

Run all automated tests:

```
php artisan test
```

Run the page crawler on demand (also runs nightly at 2am):

```
php artisan app:crawl-routes
```

Check scheduled tasks are each listed once:

```
php artisan schedule:list
```

Tip: run `php artisan test` before and after any big change. Green means the core of the app still works; a red test points straight at what broke.

What's left

The testing sweep and the deferred bug list are **complete**. One major initiative remains, to be tackled when you're ready:

- **Go-live planning** — hosting, patient-data security (DPDP compliance), and automated backups before the clinic runs on it for real.